

AUSTIN EIDE

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EDUCATION

University of Nebraska-Lincoln

August 2023

Ph.D. in Mathematics

Advisor: Xavier Pérez Giménez

Dissertation: *Aspects of the averaging process*

Colorado College

May 2016

B.A. in Mathematics, with Distinction

EXPERIENCE

Pacific Northwest National Laboratory

Jul 2026 - Present

Postdoctoral Research Associate

Seattle, WA, USA

Eldridge Capital Management

Oct 2025 - May 2026

Associate

Des Moines, IA, USA

Toronto Metropolitan University

Aug 2023 - Aug 2025

Postdoctoral Fellow

Toronto, ON, Canada

Los Alamos National Laboratory

May 2022 - Aug 2022

NSF MSGI Intern

Los Alamos, NM, USA

University of Nebraska-Lincoln

Aug 2017 - Jul 2023

Graduate Teaching Assistant

Lincoln, NE, USA

RESEARCH INTERESTS

Random graphs, stochastic processes, Markov chains, randomized algorithms, network science, asymptotic enumeration, probabilistic combinatorics, analytic combinatorics, probability theory, explainable machine learning.

RESEARCH

7. **Direct paths in the temporal hypercube** (with P. Prałat & M. Gössgens) 2025
Preprint, *arXiv:2509.18931*. (Submitted.)
6. **Multiset metric dimension of Binomial Random Graphs** (with P. Prałat) 2025
Preprint, *arXiv:2507.11686*. (Submitted.)
5. **Playing Sudoku on random 3-regular graphs** (with J. Dippel, P. Prałat, & D. Willhalm) 2025
Preprint, *arXiv:2503.07335*. (Submitted.)
4. **Linear Colorings of the Binomial Random Graph** (with P. Prałat) 2025
Discrete Mathematics, Vol. 348, No. 2 (2025): Article 114278.
3. **Burning Random Trees** (with L. Devroye & P. Prałat) 2025
Electronic Communications in Probability, Vol. 30, No. 14 (2025): 1-9.
2. **Concentration and CLT for the Averaging Process on $\mathbb{Z}^d$** 2024
Preprint, *arXiv:2404.02351*. (Submitted.)
1. ***Aspects of the Averaging Process*** 2023
Ph.D. Thesis, University of Nebraska—Lincoln.

TECHNICAL SKILLS

Programming: Python (TensorFlow, PyTorch, Scikit-learn, NumPy, SciPy, PyMC, pandas, xarray), R, HTML, Git

Modeling: neural networks, random forests, linear & logistic regression, clustering, MCMC

SELECTED PRESENTATIONS

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| 10. CanaDAM 2025
“Playing Sudoku on random 3-regular graphs” (invited) | <i>May 2025</i> |
| 9. Analysis of Algorithms 2025
“Playing Sudoku on random 3-regular graphs” | <i>May 2025</i> |
| 8. University of Toronto Probability Seminar
“Random averages on the integer lattice” | <i>April 2025</i> |
| 7. Graphs@TMU Seminar
“Random averages on $\mathbb{Z}^d$ ” | <i>February 2025</i> |
| 6. Graphs@TMU Seminar
“Burning Random Trees” | <i>October 2024</i> |
| 5. UNL Discrete Math Seminar
“Linear colorings of the binomial random graph” | <i>January 2024</i> |
| 4. Graphs@TMU Seminar
“The averaging process for finite graphs” | <i>October 2023</i> |
| 3. NSF Mathematical Sciences Summer Research Symposium
“An explainable CNN model for predicting the Southern Annular Mode” | <i>August 2022</i> |
| 2. Drake University
“The Mathematics of Gerrymandering” | <i>November 2018</i> |
| 1. UNL Discrete Math Seminar
“Sampling and Graph Problems in Political Redistricting” | <i>September 2018</i> |

TEACHING

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| 1. Math in the City (Senior Capstone, TA, UNL) | <i>Fall 2019, Spring 2020, Fall 2022, Spring 2023</i> |
| 2. Linear Algebra* (UNL) | <i>Fall 2022</i> |
| 3. Differential Equations* (UNL) | <i>Spring 2021, Spring 2022</i> |
| 4. Math for Elementary Educators* (UNL) | <i>Fall 2021</i> |
| 5. Calculus I* (UNL) | <i>Fall 2020</i> |
| 6. Calculus II (TA, UNL) | <i>Fall 2017, Spring 2018</i> |
| 7. College Algebra* (UNL) | <i>Fall 2018, Spring 2019</i> |

(*Indicates instructor of record.)

AWARDS, HONORS, AND FELLOWSHIPS

TMU Faculty of Science Travel Award (TMU) A \$1500 award for postdoctoral research travel.	<i>April 2025</i>
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Grace Chisolm & William Henry Young Award (UNL)

November 2019

Awarded to a graduate student for excellence in scholarship by the Department of Mathematics.

Othmer Fellowship (UNL)

August 2017

A prestigious, university-wide fellowship providing \$8,000 in annual support for graduate research.

Participant, Graduate Research Workshop in Combinatorics

June 2021

Worked with multiple teams on problems in combinatorics and graph theory.

Graduate Fellow, Voting Rights Data Institute

June 2018 - August 2018

Program on the mathematics of political redistricting hosted jointly at Tufts, MIT, and Harvard.

Graduated *magna cum laude* (Colorado College)

May 2016

Phi Beta Kappa (Colorado College)

Inducted May 2016

MENTORING AND SERVICE

UNL Directed Reading Program

2019-2023

*Co-Founder and Organizer**University of Nebraska—Lincoln*

- Co-founded and managed the Directed Reading Program at UNL, pairing graduate student mentors with undergraduates for semester-long guided reading projects.

Coordinated Differential Equations

2023

*Co-Editor**University of Nebraska—Lincoln*

- Collaborated with Dr. Adam Larios to adapt an open-source Differential Equations textbook for use at UNL.

UNL Graduate Student Assembly

2020-2023

*Math Department Representative and Communications Chair**University of Nebraska—Lincoln*

- Led communications for UNL's Graduate Student Assembly.
- Developed and presented new legislation aimed at improving graduate student life at UNL.

Great Plains Alliance

2017-2023

*Presenter and Ambassador**University of Nebraska—Lincoln*

- Gave talks and discussed graduate school opportunities at regional institutions.

OTHER PROFESSIONAL EXPERIENCE

Colorado College

Aug 2016 - May 2017

*Mathematics Paraprofessional**Colorado Springs, CO, USA*

- Served as TA for undergraduate mathematics courses.
- Assisted in department administrative duties.